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RESEARCH

Spatio-temporal analysis of surface urban heat island and thermal variability in relation to LULC patterns: a case study of Delhi, India

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Abstract
Urbanization and climate change are intertwined with each other, which has altered the local climate and resulted in the formation of urban heat island (UHI). This impacts the environment and human well-being due to increased heat stress, which may worsen during extreme weather events. The study aims to investigate the changes in the spatiotemporal pattern of land use/land cover, land surface temperature (LST), UHI, and thermal comfort in Delhi and its four major neighbouring cities. The study has used Landsat images for assessing LST, UHI, and urban thermal field variance index (UTFVI) for the years 2010, 2019, 2020, 2021, and 2022. The results showed an increase in the area of vegetation, urban, and barren land by 271km^2, 214km^2, and 130km^2, respectively, during 2010–2020. The study also revealed an increase in mean LST by 2°C, indicating a warming trend from 2010 to 2019. However, prolonged COVID-19 lockdown and reduced anthropogenic activities resulted in a 4 °C reduction in mean LST in 2021 and an increase in thermal comfort. The mean LST difference between vegetation and urban areas was about 4–5 °C. In 2010, the maximum area was within the UHI range of 0.5–0.6, which has shifted to a higher range of > 0.7 in 2019 and 2022. During the heatwave condition, an increase in the heat-exposed area is observed.

Keywords Land-use/land-cover · Land surface temperature · Urban heat island · Urban thermal field variance index · Delhi

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1 Introduction

Climate change is defined as long-term alterations in weather and temperature patterns that have a direct influence on regional and local climatic conditions (UN 2023). There are many factors that contribute to climate change, both natural and artificial, with urbanization being one of the main factors (NRDC 2021). In 2022, the global population had reached 8 billion, with more than half of those people living in cities, and it is expected that 2.5 billion more people will live in urban areas by 2050 (SDG 2023; WUP 2018). Cities are the engines of economic growth, accounting for 80% of the world's gross domestic product. It provides opportunities for employment in the service and industry sectors and better education, which fuels urbanization (WUP 2018). Similar to this, the study region, which includes Delhi and its four main neighbouring cities (Ghaziabad, Faridabad, Gurugram, and Gautam Buddha Nagar), has evolved due to economic growth and population pressure. Anthropogenic activities greatly influence the region, resulting in a highly heterogeneous urban surface that has altered the land use/land cover (LULC) pattern and increased its vulnerability to extreme events due to climate change. LULC is a fundamental indicator of human-environment interaction. Land cover represents the natural features on the Earth's surface, while land use describes how humans utilize land for various purposes.

Urban expansion has caused a decrease in vegetation and agricultural area, a decline in ecosystem services, biodiversity loss, higher utilisation of natural resources, and a rise in natural disasters, making cities more vulnerable and threatened (Kumar et al. 2019; Koko et al. 2020; WCR 2022; Tabassum 2024). It is important to assess the ratio of natural cover to urban cover through continuous monitoring of LULC dynamics because the conversion of natural porous surfaces to artificial non-porous surfaces has become a major concern at the national and global levels (Keshtkar and Voigt 2016; Chanu et al. 2021; Sattari et al. 2024). This has modified the microclimate of urban areas by reducing evapotranspiration and moisture content, increasing solar radiation absorption, altering wind turbulence, precipitation, and albedo, shrinking ecological space, rising waste heat, and elevating land surface temperature (LST) (Stewart and Oke 2012; Madanian et al. 2018; Mohammad et al. 2019; Khan et al. 2023). Transformations in LULC have a close relationship with LST, which refers to the skin temperature of the radiative surface of Earth, differing from surface air temperature above the ground (Wan 1999; Sahoo et al. 2022). LST is an important environmental indicator, as it influences the surface energy budget (Oke 1982; Tiwari and Kanchan 2024). The increase in LST caused by urbanization has disrupted various biophysical processes (such as photosynthesis, plant phenology, and evapotranspiration), air quality, destabilized ecosystems, impacted public health, and favoured urban heat island (UHI) formation (Zhi et al. 2020; Jiang et al. 2023).

The UHI is a climatic phenomenon caused by temperature differences; urban areas often experience higher temperatures than non-urban areas (Voogt and Oke 2003; Sahoo et al. 2022). This temperature difference arises from distinct surface and atmospheric processes, urban areas dominated by impervious surfaces (high heat capacity and low albedo), greater absorption and trapping of heat in built structures, anthropogenic heat emissions from vehicles and industries, lower wind speeds, higher concentrations of pollutants, reduced vegetation, and reduced evaporative cooling, all of which collectively contribute to the urban warming effect. In contrast, non-urban areas with natural vegetation and waterbodies that have greater albedo and higher evapotranspiration remain relatively cooler due to the

natural cooling effect (Mohammad et al. 2019; Das et al. 2020; Garcia 2022; Gohain et al. 2023). The presence of UHI affects the thermal balance that further increases the use of air conditioners, electricity consumption, and greenhouse gas emissions (Ahmed 2018; Li et al. 2018; Tiwari and Kanchan 2024; Zargari et al. 2024). This emphasizes the need to determine the UHI pattern for creating a sustainable action plan for human well-being.

The impact of UHI is exacerbated due to heatwave conditions, as it increases heat stress which may cause illnesses such as heat stroke, heat exhaustion, sunburn, and dehydration that are life-threatening (IPCC 2022; Pan et al. 2024). A heatwave, as defined by the Intergovernmental Panel on Climate Change, " is a period of abnormally hot weather, often defined with reference to a relative temperature threshold, lasting from two days to months" (IPCC 2022) while India Meteorological Department defines heatwave as a condition when the actual maximum air temperature reaches ≥ 45 °C for at least two consecutive days in an area (IMD 2022). The intensity, frequency, and duration of heatwaves have increased due to human-induced climatic changes resulting from urban development and anthropogenic activities, which account for 37% of heat-related fatalities (Vicedo-Cabrera et al. 2021; IPCC 2022). In India, 6,167 heat-related fatalities have been recorded for the period of 2010 and 2018 (IS 2019). Therefore, estimating the thermal comfort level and analysing heat-exposed areas during heatwave condition is crucial.

There are numerous studies that show a relationship between LST, UHI, and spectral indices (normalized difference built-up index and normalized difference vegetation index) in Delhi (Mohan et al. 2012; Kumari et al. 2018; Suhail et al. 2019; Singh et al. 2022a, b; Bala and Dar 2024) and comparative studies between metropolitan cities (such as Delhi, Mumbai, and Kolkata) (Grover and Singh 2015; Sultana and Satyanarayana 2019; Jallu et al. 2022). These previous studies have primarily focused on the Delhi region. However, neighbouring cities are undergoing significant population pressure and expanding over time, so it becomes important to explore the changes in neighbouring cities as well. Therefore, this study makes an attempt to analyse multiple time periods, allowing the detection of both decadal trends and interannual variations, including conditions influenced by the COVID-19 lockdown and the occurrence of heatwave. The main objectives of this study are: (1) to examine the LULC changes from 2010 to 2020; (2) to retrieve LST for the years 2010, 2019, 2020, 2021, and 2022, as well as establish a relationship with LULC; and (3) to investigate the UHI_{index} and urban thermal area variance index (explained in Sect. 3.4 and 3.5) for assessing thermal behaviour. The outcome of the study will be helpful for city planners and policymakers to understand the effect of UHI on the environment and human well-being, which can facilitate the creation of sustainable cities by formulating better policies for urban management and UHI mitigation.

2 Study area

Delhi, the capital city of India, along with its four major surrounding cities - Ghaziabad, Faridabad, Gurugram, and Gautam Buddha Nagar - serves as an economic hub of India and is located between longitudes 77°21′12.36"E to 77°25′16.78"E and 28°14′15.59"N to 28°55′52.05"N latitudes (Fig. 1). It spreads over an area of 5955 km^2. The area lies in the Cwa climatic zone, which is characterized by a humid subtropical climate with hot summers and dry winters under the Köppen-Geiger climate classification.

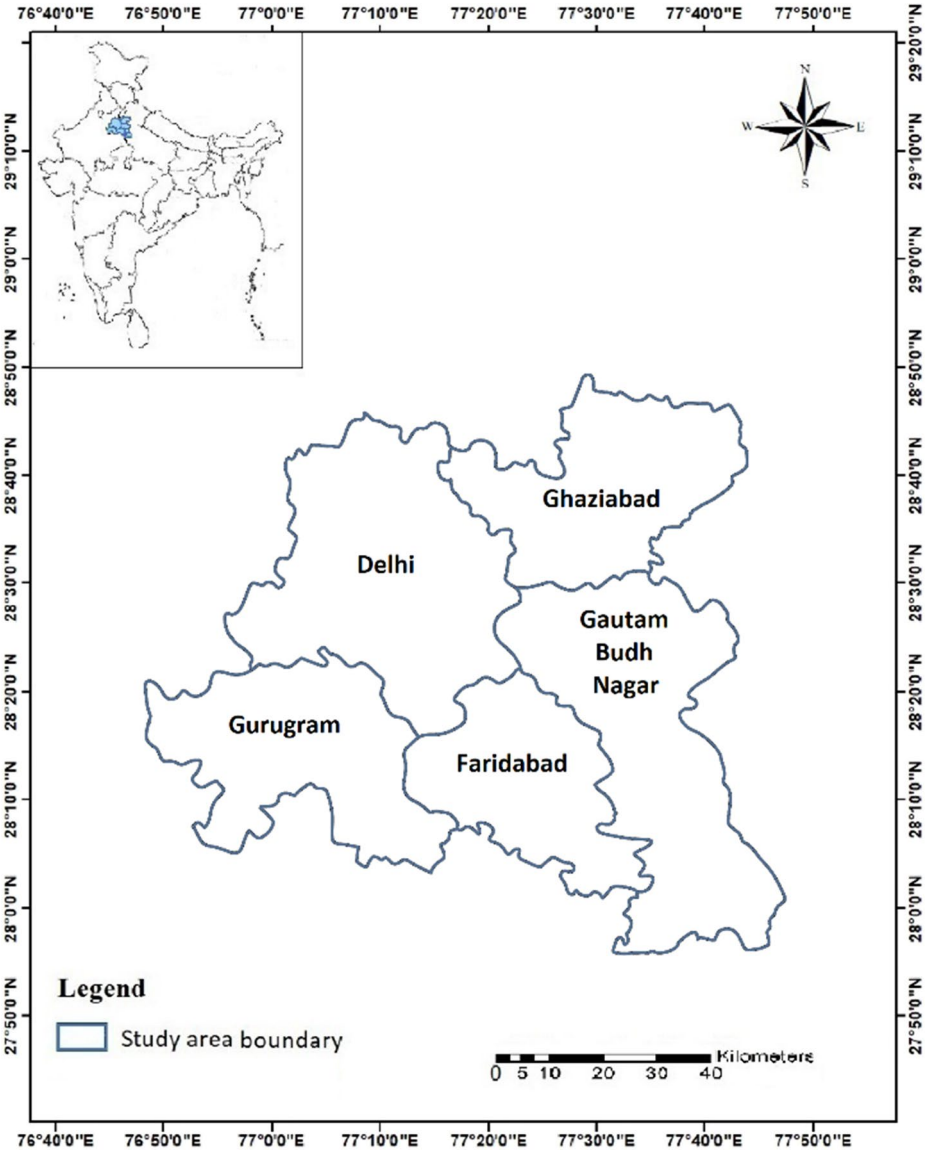

Fig. 1 Location map of the study area

3 Materials and methods

3.1 Satellite data

For this work, Landsat satellite images were freely downloaded from the Earth Explorer website of the USGS (https://earthexplorer.usgs.gov/) and used to evaluate thermal changes in the study area (Table 1). These images were acquired for the month of May, this time is

Table 1 Details of data used in the study

Satellite	Sensor	Year	Acquisition date	Path/Row	Cloud Cover (%)	Resolution (m)
Landsat 5	TM	2010	21-05−2010	146/40	0	30
			12-05−2010	147/40	0	
Landsat 8	OLI/TIRS	2019	30-05−2019	146/40	0.01	30
			05-05−2019	147/40	0	
		2020	16-05−2020	146/40	0.03	30
			07-05−2020	147/40	0.01	
		2021	−2021	146/40	7.35	30
			26-05−2021	147/40	0	
Landsat 8/9	OLI/TIRS OLI-2/IRS-2	2022	14-05−2022*	146/40	0.02	30
			13-05−2022	147/40	0	

*Heatwave event occurred

the peak heat period, characterized by frequent and severe heatwave occurrences in Delhi. In the study, we compared the LST, UHI, and UTFVI data for different years in two ways: (i) decadal change comparison (2010–2020), and (ii) comparison based on four consecutive years, that include non-heatwave day (2019), COVID lockdown (2020), partial prolonged COVID lockdown (2021), and heatwave day (2022). The study has a limitation that for analysing the difference between a non-heatwave day (non-HW) and a heatwave day (HW), single date data from Landsat was available when heatwaves occurred. Despite this limitation, the study's findings provide significant insightful information about the relationship of UHI and heatwave conditions.

3.2 LULC classification

The foremost step is image pre-processing, which includes creating a single multi-band image, subsetting the area of interest, and image enhancement that helps in observing biophysical features in the image more accurately (Alawamy et al. 2020). For the present study, Landsat images for the years 2010 and 2020 were used and classified into six different LULC classes, namely vegetation, agricultural area, urban area, rural area, barren land, and waterbody, using supervised classification with maximum likelihood algorithm (Singh et al. 2022a, b). The accuracy assessment was computed by calculating overall accuracy and kappa statistics using 610 random points and reference data (high-resolution Google Earth data and field visits) to validate the accuracy of these classified LULC maps (Sarkar and Chouhan 2019). The results are presented in Table 2, which includes an estimation of area in each category and an accuracy assessment for the years 2010 and 2020 (data has been derived and modified from Singh et al. 2022a, b).

3.3 Retrieval of LST

LST is essential for understanding the thermal characteristics of the Earth's surface. The LST is derived from the thermal bands of satellite images (band 6 in Landsat 5 and 7, band 10 in Landsat 8 and 9) using the Landsat data user handbook approach (USGS 2019a; USGS 2019b). The steps for LST retrieval are as follows:

Table 2 Estimation of LULC changes and accuracy assessment for 2010 and 2020, modified from Singh et al. (2022a, b)

LULC categories	Area				Difference
	2010		2020		2010–2020
	km^2	%	km^2	%	%
Vegetation	1168.88	19.63	1440.36	24.19	4.56
Urban	993.53	16.68	1207.59	20.28	3.6
Rural	196.31	3.3	218.22	3.66	0.36
Agricultural area	2452.3	41.18	1856.82	31.18	−10
Waterbody	132.32	2.22	89.48	1.5	−0.72
Barren land	1012.17	17	1143.04	19.19	2.19
Total	5955.51		5955.51		
Overall accuracy	93%		89%		
Kappa coefficient	0.91		0.87		

(i) The fundamental step is to convert the digital number (DN) of the thermal band into spectral radiance (L_λ) using equations (1a) and (1b) (Abir and Saha 2021; Salam et al. 2023). The equation requires radiance scaling factors, which are available in the metadata file of the satellite image.

For Landsat 5 and 7,

$$L_\lambda = \left[\frac{L_{\lambda MAX} - L_{\lambda MIN}}{Q_{CALMAX} - Q_{CALMIN}}\right] \times (Q_{CAL} - Q_{CALMIN}) + L_{\lambda MIN} \tag{1a}$$

where,

L_λ is spectral radiance at sensor in Watts/(m^2 sr µm),

$L_{\lambda MAX}$ is spectral radiance which is scaled to Q_{CALMAX} in Watts/(m^2 sr µm),

$L_{\lambda MIN}$ is spectral radiance which is scaled to Q_{CALMIN} in Watts/(m^2 sr µm),

Q_{CAL} is quantized calibrated pixel value of DN,

Q_{CALMAX} is maximum quantized calibrated pixel value of DN (corresponding to $L_{\lambda MAX}$), and

Q_{CALMIN} is minimum quantized calibrated pixel value of DN (corresponding to $L_{\lambda MIN}$).

For Landsat 8 and 9,

$$L_\lambda = M_L * Q_{CAL} + A_L \tag{1b}$$

where,

M_L is radiance multi-band,

A_L is radiance add band, and

Q_{CAL} is quantized calibrated pixel value in DN.

(ii) The next step is to calculate brightness temperature (BT) from spectral radiance using the thermal constants mentioned in the metadata file. The conversion formula is given in Eq. (2), and the estimated BT is in Kelvin, which is required to convert to Celsius by subtracting 273.15 (Jallu et al. 2022).

$$BT = \frac{K_2}{\ln\left(\frac{K_1}{L_\lambda} + 1\right)} - 273.15 \tag{2}$$

where,
BT is effective satellite brightness temperature in Celsius,
K_1 is calibration constant 1,
K_2 is calibration constant 2, and
L_λ is spectral radiance.

(iii) Then, the land surface emissivity (LSE) is measured using the normalized difference vegetation index (NDVI) threshold method. There are several surface factors that influence emissivity as a function of wavelength, such as water content, chemical composition, texture, and structure (Das et al. 2020). To calculate emissivity (ε), the proportion of vegetation (P_V) is calculated from the minimum and maximum NDVI values using the following Eqs. (3–5) (Mujabar and Rao 2018; Abir and Saha 2021):

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{3}$$

where,
NIR is near-infrared band, and RED is red band.

$$P_V = \left(\frac{NDVI - NDVI_{MIN}}{NDVI_{MAX} - NDVI_{MIN}}\right)^2 \tag{4}$$

where,
NDVI is Normalize Difference Vegetation Index,
$NDVI_{max}$ is maximum DN value from NDVI image, and
$NDVI_{min}$ minimum DN value from NDVI image.

$$\epsilon = 0.004P_V + 0.986 \tag{5}$$

where,
ε is land surface emissivity, and P_v is proportion of vegetation.

(iv) After the estimation of LSE, the following Eq. (6) is used to calculate LST (Ahmed 2018; Gohain et al. 2023):

$$LST(^\circ C) = \left(\frac{BT}{1 + \left(\frac{\lambda * BT}{\rho}\right) * \ln(\varepsilon)}\right) \tag{6}$$

where,
BT is brightness temperature in Celsius,
λ is effective wavelength,
ε is emissivity,

ρ is constant ($\rho = h\frac{c}{\sigma} = 1.438 * 10^{-2}\ mK$),
σ is Boltzmann constant (1.38×10^{-23} J/K),
h is Planck's constant (6.626×10^{-34} J S), and
c is velocity of light (2.998×10^{8} m/s).

3.4 Urban heat Island index (UHI_{index})

The UHI_{index} quantifies the temperature difference between urban and non-urban areas. To understand the UHI pattern, the UHI_{index} is computed, and variations in the UHI are determined from the LST pattern. The UHI_{index} ranges from 0 (no UHI effect) to 1 (UHI effect). The UHI_{index} is calculated using Eq. (7) as follows (Sahoo et al. 2022; Zargari et al. 2024):

$$UHI_{index} = \frac{LST - LST_{min}}{LST_{max} - LST_{min}} \tag{7}$$

where,
LST is land surface temperature,
LST_{max} is maximum LST value, and
LST_{min} is minimum LST value.

UHI is divided into 7 classes (<0.2, 0.2–0.3, 0.3–0.4, 0.4–0.5, 0.5–0.6, 0.6–0.7, >0.7) in the rising intensity order of UHI, and the area covered by each UHI class is calculated (Sahoo et al. 2022). Hot spots and cold spots are identified based on the pixels with the maximum or minimum UHI_{index}, and it is also illustrated how LULC affects the UHI_{index} using Google Earth images (Fig. 2).

3.5 Urban thermal field variance index (UTFVI)

UTFVI is used to analyse the effects of UHI, such as thermal comfort level, extent of heat-exposed areas, ecological disturbances, and quality of life in the area (Gohain et al. 2023). For example, a higher LST will result in greater heat effects and weaker environmental

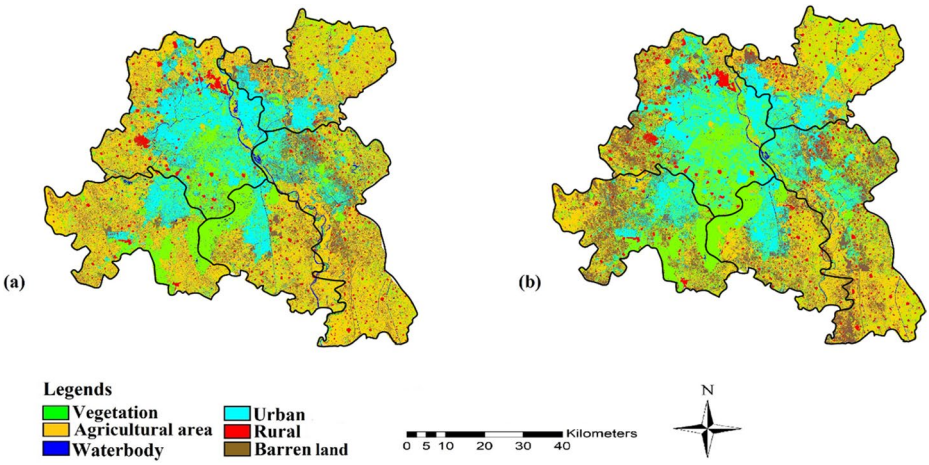

Fig. 2 Classified LULC maps of the study area of (**a**) 2010 and (**b**) 2020

health. It is categorized into six different ecological evaluation zones (EEI) or thermal comfort levels, as given in Table 6 (Renard et al. 2019). It is computed using Eq. (8) given below (Mokarram et al. 2024):

$$UTFVI = \frac{LST - LST_m}{LST_m} \tag{8}$$

where,
LST is land surface temperature, and
LST_m is mean LST.

4 Results and discussion

4.1 Spatio-temporal analysis of LULC

The classified LULC maps have shown considerable changes in the six distinct LULC classes of the study area during the period from 2010 to 2020 (Table 2; Fig. 3, derived and modified from Singh et al. 2022a, b). It is revealed that urban area has continuously increased and has reached from 993.53 km^2 in 2010 to 1207.59 km^2 in 2020 due to the systematic shifting of vegetation and agricultural area to barren land and then to urban area. The pattern of urban expansion varies across the study area, which includes core-to-periphery growth in Delhi, periphery-to-semi-radial growth in Gurugram, longitudinal growth in Faridabad, northern-to-southern growth in Gautam Buddha Nagar, and ribbon growth in Ghaziabad. The study area is under immense demographic pressure (Fig. 4) because there are many job opportunities from various government and private sectors, as the region is an industrial and commercial hub, with the introduction of new economic policy (1991), the entry of multinational companies, and the creation of Special Economic Zones in Noida and

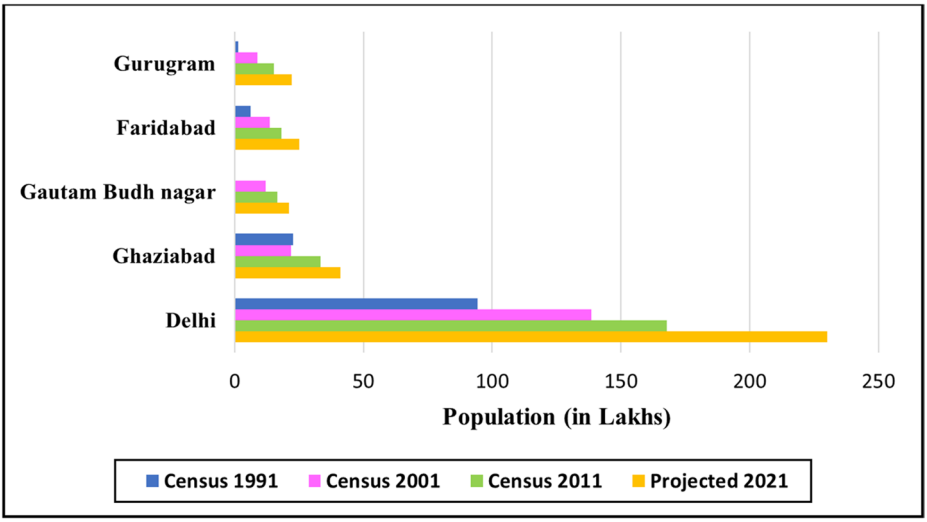

Fig. 3 Decadal population growth (Source – Census 1991, 2001, 2011)

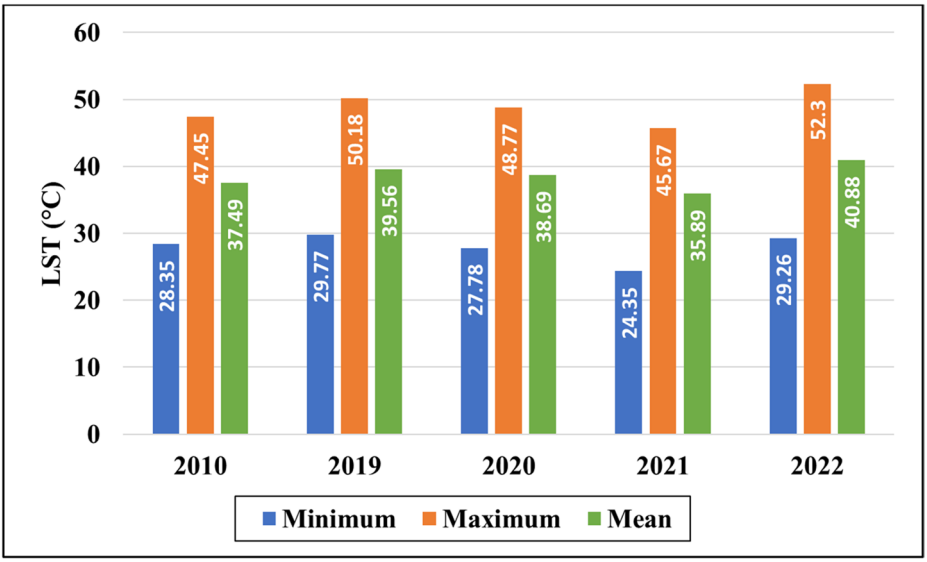

Fig. 4 LST variation during May month over the study area

Gurugram. The inflow of people leads to the expansion of existing residential, industrial, and commercial areas, the establishment of new ones, the formation of the new district Gautam Buddha Nagar, and the development of infrastructure projects that provide facilities and good connectivity (such as metro rail and highways) among all 5 regions of the study area (Census 2011; DES 2019; NCRPB 2021; Metro 2022; Singh et al. 2022a, b). However, urban expansion has negatively affected natural environmental conditions such as vegetation, microclimate, warming, increased impervious surfaces, and pollution.

Vegetation had been adversely affected previously due to the degradation of vegetation at mining sites in the Aravali range of the study area, their transformation into urban and agricultural areas, and illegal encroachment (Singh et al. 2022a, b). The creation of biodiversity parks, restriction on mining, large-scale plantation and restoration activities led by various organisation's initiatives have restored the depleted vegetation, increasing it by 4.56% during 2010–2020. It is also revealed that the agricultural area is reduced from 41.18% in 2010 to 31.18% in 2020 as the area is changing into urban or barren land. This has been caused by the shift in the economic base from agriculture to industry and commerce, which has resulted in the acquisition of more land for settlement and developmental projects for the working population.

The area of waterbodies (river, lake, wetland, and canal) diminished from 2010 (132.32 km^2) to 2020 (89.48 km^2) due to encroachment and their infilling for agricultural and urban construction uses. In order to rejuvenate and conserve the existing waterbodies, there is a need to adapt water management techniques. The significant change was the gradual filling of the abandoned mining pits with water, which transformed them into thriving lakes. Barren land (bare soil, vacant land, fallow land, and exposed rocky areas) is increased due to the degradation of land, and land is acquired by the government and private developers for townships. During 2010, the total rural area was 196.31 km^2, which further increased

to 218.22 km^2 during 2020 as the rural population expanded along with the rise in village size and number.

4.2 Spatio-temporal distribution of LST

The spatio-temporal pattern of LST over Delhi and its surrounding four major cities is presented for the month of May in the years 2010 (decadal change), 2019 (non-HW), 2020 (COVID lockdown), 2021 (prolonged partial lockdown), and 2022 (HW). Figure 5 illustrates the maximum, minimum, and mean LST for each year considered. The mean LST has gradually increased from 37.49 °C (2010) to 39.56 °C (2019); i.e., the mean temperature has increased by 2.07 °C due to changes in the urban build-up environment, increased anthropogenic activities, and air pollution. This difference in the mean temperature is further raised by 1.32 °C in 2022 (HW) when compared to 2019 (non-HW), due to the occurrence of a heatwave. This indicates that heatwave events could trigger the LST. While the mean LST is significantly reduced by approximately 2 °C and 4 °C in 2020 and 2021, respectively, in comparison to 2019 and 2022, due to the COVID-19 lockdown situation, restricted human activities, reduced air pollution levels, and pleasant weather conditions, which include increased frequency of western disturbances (Singh et al. 2024). As this phase finished off, a spike in LST was again observed in 2022.

Further, the LST values are categorised into six different temperature zones, and their extent is computed (Table 3; Fig. 6), where red color indicates higher temperatures and blue color indicates lower temperatures. It is observed that areas from lower temperature zones have shifted to higher temperature zones, which indicates that the minimum and maximum LST ranges have been raised. During 2010–2022, the area under 35 °C decreased from 152.42 km^2 to 73.55 km^2, whereas the area above 44 °C increased from 1310.10 km^2 to 2163.06 km^2. It is noticed that the majority of the area was under the 42–44 °C temperature

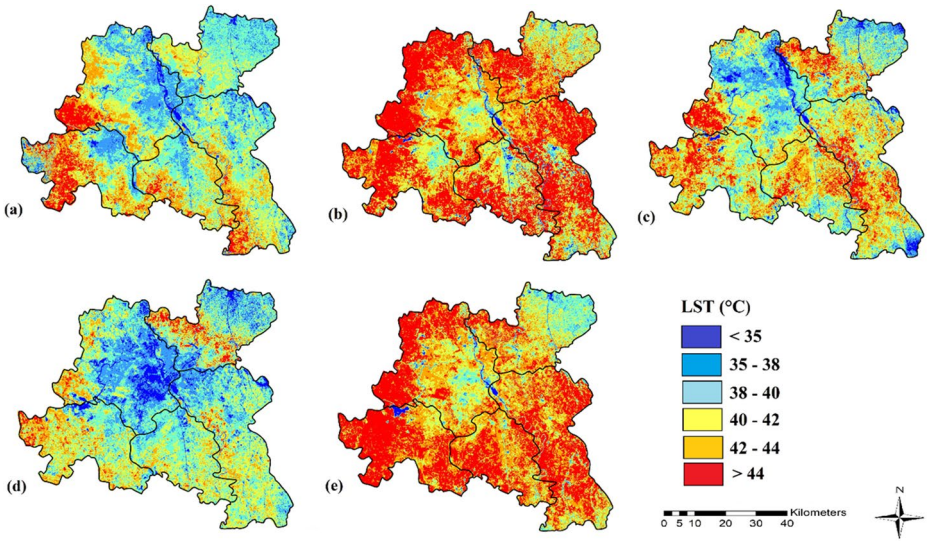

Fig. 5 Spatio-temporal distribution of LST in the summer season: **(a)** 2010, **(b)** 2019, **(c)** 2020, **(d)** 2021 and **(e)** 2022

Table 3 Changes in the area under each LST range

Year	LST range (°C) with their respective areas in Km^2 and (percentage)					
	<35	35–38	38–40	40–42	42–44	>44
2010	152.41 (2.56)	674.50 (11.33)	872.45 (14.65)	1341.11 (22.52)	1604.91 (26.95)	1310.10 (22.00)
2019	70.85 (1.19)	278.25 (4.67)	550.84 (9.25)	1290.96 (21.68)	1623.11 (27.25)	2141.48 (35.96)
2020	165.33 (2.78)	659.40 (11.07)	1272.52 (21.37)	1635.76 (27.47)	1167.68 (19.61)	1054.79 (17.71)
2021	284.62 (4.78)	807.54 (13.56)	1746.74 (29.33)	1834.65 (30.81)	739.47 (12.42)	542.48 (9.11)
2022	73.55 (1.24)	271.46 (4.56)	563.58 (9.46)	1167.03 (19.60)	1716.80 (28.83)	2163.06 (36.32)

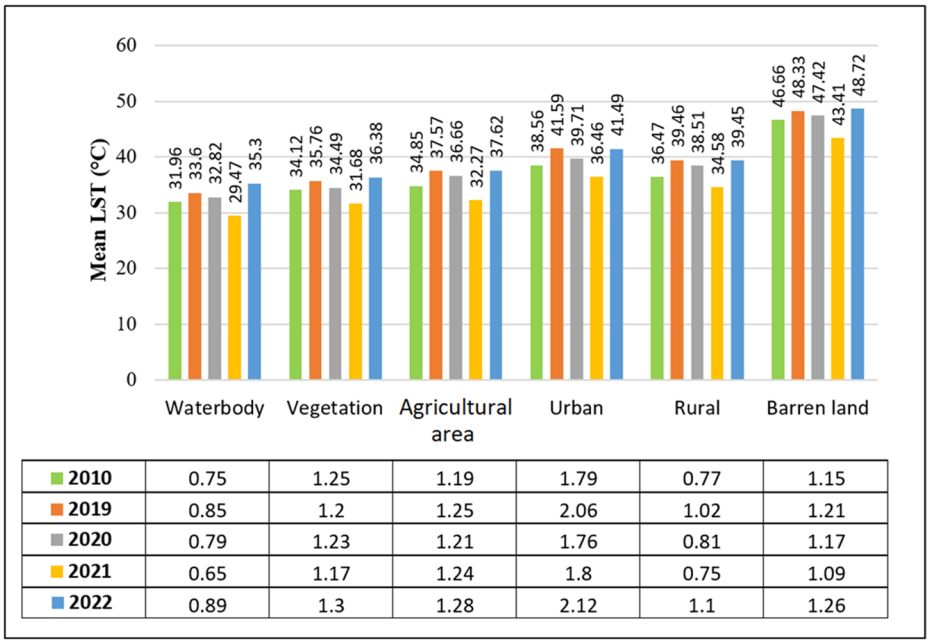

Fig. 6 LULC class-wise mean LST (°C) and standard deviation for the month of May in different years

zone during 2010, 2019, and 2022; however, it shifted to 38–40 °C and 40–42 °C zones during 2020 and 2021, indicating the movement from higher to lower LST zones. The results also revealed that areas below 35 °C increased while areas above 44 °C decreased in these LST zones during 2020 and 2021. The area under the higher LST range, i.e., 42–44 °C and above 44 °C, increased during HW as compared to non-HW.

The IPCC's sixth assessment report (2022) states that the frequency and intensity of hot days and nights have increased globally. The LST maps of the study area show heat variation, which is evident in the reduction in the area under low temperature and the increment in the area under high temperature with time. This is due to urban expansion, anthropogenic heat discharge, industrialization, pollution, reduced vegetation cover, and extreme

heat events. Moreover, the imbalance in the ratio of impervious surfaces to green cover has a negative impact on the heat transfer process that eventually triggers the UHI effect, which becomes a serious concern for the development of sustainable urban environments. These variations in LST and LULC changes play an important role in the biological cycle of crops, heat stress, and climate change (Brogno et al. 2025).

4.3 LST distribution pattern for different LULC classes

There is a direct association between the changing nature of LST and changes in LULC types, as different compositions of LULC types interact differently with the sun's radiation. It is important to investigate these changes and their extent in order to understand the urban thermal environment, its impact on ecosystems, and the lives of urban dwellers. The mean LST of each LULC type for different years during the summer season is depicted in Fig. 7. It is revealed that waterbodies possess the lowest mean LST, while the highest mean LST is recorded over barren land. Vegetation and agricultural areas showed lower mean LSTs, which varied from 31.68 °C to 37.62 °C during the study period.

The mean LST difference between vegetation and urban areas was about 4–5 °C, and between waterbodies and urban areas, it was approximately 6–8 °C. The lowest LST is observed over the waterbody in Okhla Bird Sanctuary, Noida. Areas that are near the waterbodies are cooler, as they have a moderating effect on climate (USGS 2018). However, waterbodies have high heat capacity and may increase surrounding temperature during night or seasonal transitions (Steeneveld et al. 2014). Moreover, waterbodies can also increase relative humidity, which may increase heat stress (Deng et al. 2022). Thus, waterbodies mitigate heat during the daytime, while they also contribute to nighttime warming and humidity-related discomfort. The rise in temperature is observed at the Najafgarh drain, which may be due to the wastewater effluent (Wilson and Worrall 2021). Vegetation cover, urban plantations, and agricultural areas exhibit low LST because plants reflect back the incoming

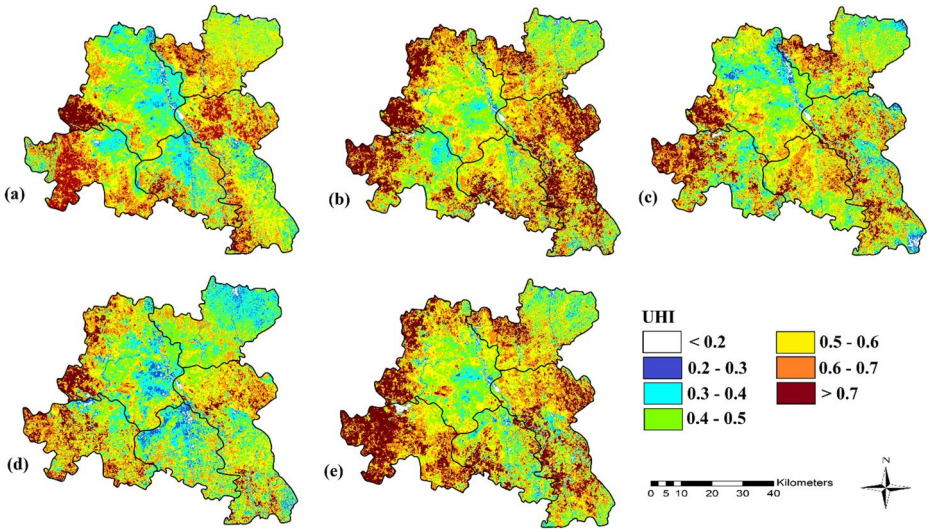

Fig. 7 Spatiotemporal variation of UHIindex : **(a)** 2010, **(b)** 2019, **(c)** 2020, **(d)** 2021 and **(e)** 2022

solar radiation, block the radiation from reaching the surface, and the evapotranspiration process cools the surrounding air (Madanian et al. 2018; Ullah et al. 2020; Tabassum 2024).

The spatial distribution and pattern of LST in urban areas (residential, commercial, and industrial) range from higher to moderate in the study area. Urban areas with high vegetation cover, like the New Delhi district, exhibit lower LST. In contrast, urban areas with scattered or less vegetation, such as Bhalswa, have a relatively higher LST. Therefore, fluctuations in LST in urban areas are noticed due to the density, size, and canopy of vegetation and their proximity to waterbody and vegetation cover. Land with pervious natural surfaces, plants, water resources, and moist soil helps to regulate LST, which affects the creation of urban heat islands.

Rural areas, characterised by natural vegetation and low anthropogenic activities, located far from urban areas, show a mean LST difference of 1 to 2 °C compared to urban areas. The temperature in rural areas close to or in the core of urban areas is the same as in urban areas.

The barren land exhibits the highest mean LST because these barren surfaces absorb more solar radiation and have low reflectivity, low soil moisture, low vegetation cover, low evapotranspiration, and poor heat dissipation, resulting in higher surface temperatures (Kara and Yavuz 2025). Elevated temperature in these areas can influence surrounding zones through heat advection.

4.4 Urban heat Island (UHI) and urban thermal field variation index (UTFVI)

The definition of the UHI by Oke describes it as the temperature difference between an urban area and its rural surroundings, primarily derived from measurements of temperature collected at ground-based monitoring stations (Oke 1982). However, the concept has been expanded to evaluate UHI using LST derived from satellite imagery, as a result of the advancement of remote sensing technology (Zhang et al. 2009). The UHI index measures the temperature difference between each pixel and finds extremes present in the study area (Schwarz et al. 2011). The maximum LST values are associated with urban surfaces, while the minimum LST values correspond to non-urban surfaces, including rural regions, vegetation, or waterbodies (Voogt and Oke 2003). Thus, this approach expands Oke's original concept into a broader and spatially continuous framework, thereby facilitating the mapping of UHI at the regional level.

The investigation of UHI intensification is now crucial as it helps in identifying the changes in local climate in the cities due to human-induced LULC changes. The UHI is the region where the LST is higher than the surrounding region's temperature, which is analysed using the UHI_{index}. The UHI images of the study area and area under different UHI intensities are shown in Fig. 8; Table 4. It is observed that there is a reduction in the area of lower UHI intensity and an increase in the area of higher UHI intensity between 2010 and 2019. This change in the UHI_{index} range is due to the conversion of natural surfaces into impervious areas. In 2010, the maximum area was under the UHI_{index} range of 0.5–0.6 (26.13%). However, it shifted to a higher range (>0.7) in 2019 and 2022, covering an area of 28.88% and 29.34%, respectively. The area under higher UHI intensity increased during the HW condition as compared to the non-HW condition. This increase in the area is due to the synergistic relationship between UHI and HW, which is driven by reduced moisture in urban areas during heatwave conditions that results in the intensification of the temperature and exacerbates the heat stress (Li and Bou-Zeid 2013; Kabisch et al. 2023; Tabassum 2024,

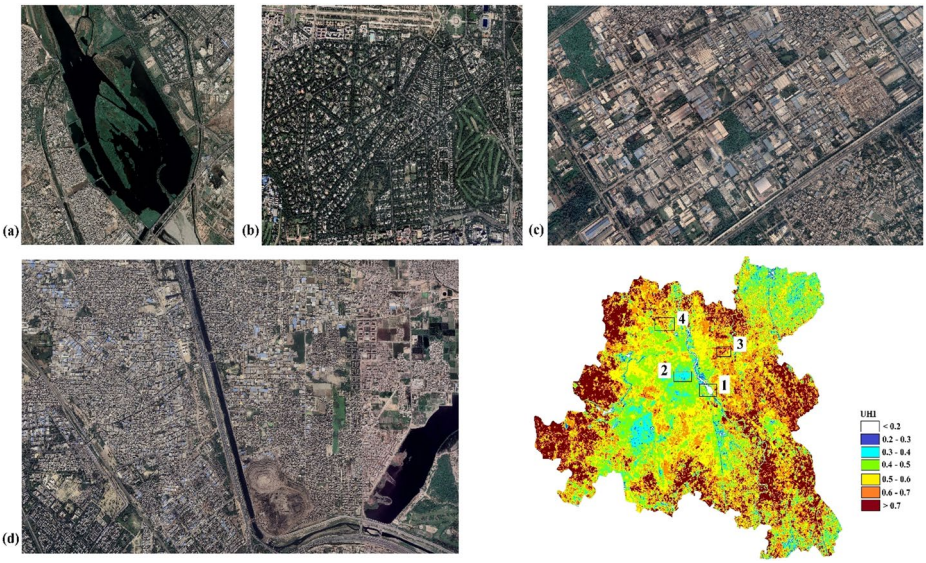

Fig. 8 High resolution Google Earth images used to illustrate the higher and lower UHI for the areas within the box in the UHIindex image of 2019: **(a)** - (1) lowest UHI over waterbody at Okhla Bird Sanctuary (Noida), **(b)** - (2) low UHI over urban area having vegetation and open spaces (New Delhi district area), **(c)** - (3) high UHI over Sahibabad industrial area and **(d)** - (4) high UHI over congested residential Bhalswa area with small industrial area between them and landfill

Table 4 Difference in the area of different UHI_{index} range in summer season

UHI_{index} range	Area (%)				
	2010	2019	2020	2021	2022
< 0.2	0.70	0.49	0.93	0.93	0.47
0.2–0.3	2.18	1.10	2.24	5.66	1.93
0.3–0.4	13.71	5.30	8.63	10.34	5.85
0.4–0.5	26.09	13.60	22.01	24.49	12.33
0.5–0.6	26.13	24.18	28.26	26.11	24.02
0.6–0.7	18.51	26.45	22.93	20.01	26.05
> 0.7	12.68	28.88	14.99	12.46	29.34

Joshi et al. 2024). During 2020 and 2021, different UHI intensity areas moved towards lower UHI_{index} ranges and noticed an increase in the area of the UHI_{index} value of less than 0.2. The shift towards the lower range is the result of reduced anthropogenic activities and pollution during the COVID-19 lockdown, as building materials and land cover remain unchanged.

The minimum UHI_{index} value identified over the waterbody and vegetation area varied between 0.2 and 0.6. The relationship between LST and LULC classes enables us to understand the significance of waterbodies and vegetation that serve as cooling agents, maintain the local climate, and contribute to the formation of urban cool islands, such as those in the New Delhi district area and Omega IV in Gautam Buddha Nagar. The high UHI is noticed over barren lands, exposed rocky regions, airports (Hindon Airport and Indira Gandhi International Airport), industrial complexes, and residential areas with high housing density or

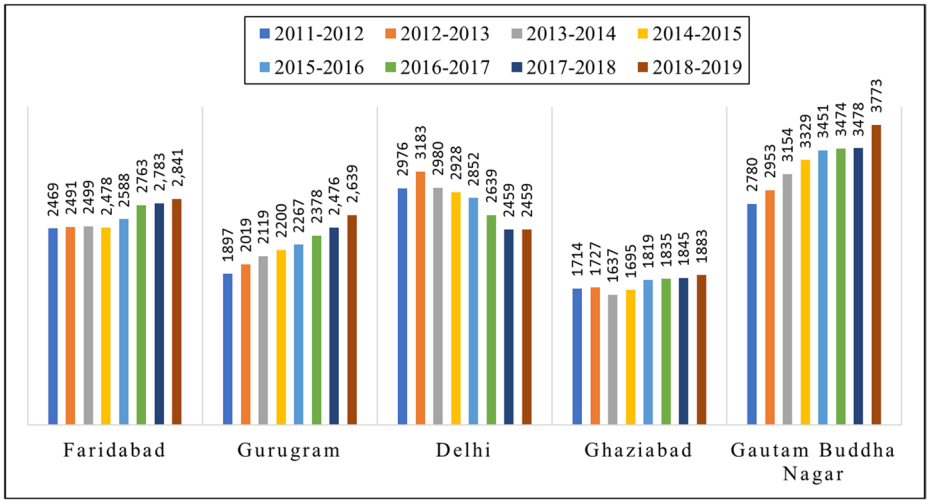

Fig. 9 Number of registered industries in the study area. Source: DES (2022), UPDES (2019), and HDES (2019)

Table 5 Number of registered vehicles in the study area

Cities	Number of newly registered vehicles				Total number of registered vehicles
	2019	2020	2021	2022	
Delhi	6,41,001	4,23,995	4,59,194	6,08,529	1,42,72,776
Ghaziabad	98,674	70,167	77,449	1,05,690	14,31,566
Gautam Buddha Nagar	91,935	65,534	77,801	1,12,986	11,14,672
Faridabad	53,620	34,826	43,828	45,698	9,89,293
Gurugram	1,29,593	75,516	68,441	96,269	15,26,137

Source: Vahan (2022)

low vegetation. The variation in the UHI_{index} values in urban areas is due to factors such as different urban geometry, building materials, anthropogenic activities, and vegetation cover (Li et al. 2018). The highest number of industries is present in Gautam Buddha Nagar (Fig. 9), and the continuous increase in the number of vehicles on the road (Table 5) may contribute to the high temperature (Jaschinsky 2024). This high-temperature trend becomes more pronounced in developing nations due to the unplanned development of cities, as there are 1731 and 321 unauthorized colonies present in Delhi and Ghaziabad, respectively (DDA 2022; GDA 2021).

UTFVI is a quantitative analysis of UHI impact that is used to determine the level of thermal comfort or discomfort zones, areas under heat stress for urban dwellers, and the ecological quality of life in urban areas (Maithani et al. 2020). The spatio-temporal distributions of the UTFVI map over the study area during the summer season for 2010, 2019, 2020, 2021, and 2022 are shown in Fig. 10.

According to Table 6, a large portion of the study area falls into two extreme categories: areas with optimal microclimate having excellent EEI or none UHI (UTFVI < 0) are either

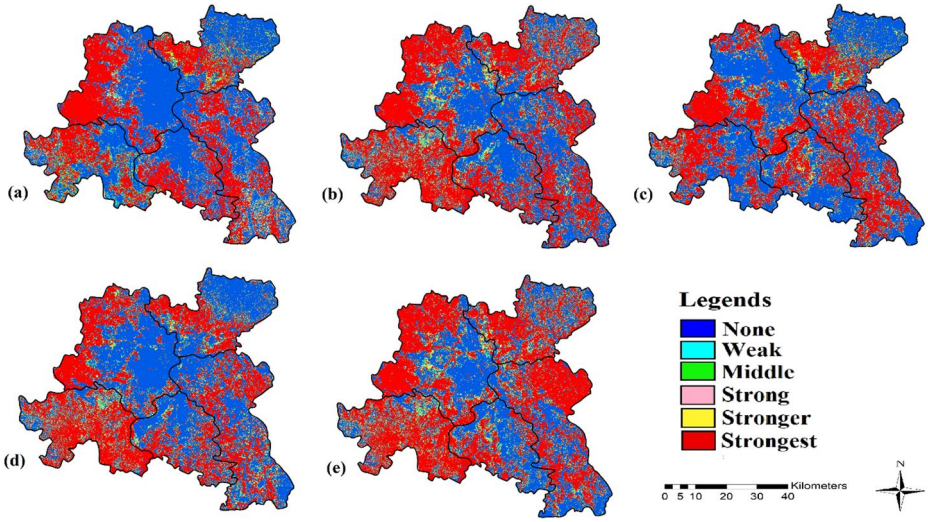

Fig. 10 Maps of the UTFVI of different years in the area: **(a)** 2010, **(b)** 2019, **(c)** 2020, **(d)** 2021 and **(e)** 2022

Table 6 Temporal variation in the area under UTFVI threshold during the study period

	Area (%)					
EEI	Excellent	Good	Normal	Bad	Worse	Worst
UHI	None	Weak	Middle	Strong	Stronger	Strongest
UTFVI	**<0**	**0–0.005**	**0.005–0.010**	**0.010–0.015**	**0.015–0.020**	**>0.020**
2010	48.43	5.39	2.50	4.10	3.22	36.37
2019	44.24	3.50	3.41	4.29	4.17	40.39
2020	49.08	3.11	3.22	3.14	3.28	38.17
2021	50.03	3.62	3.56	3.45	3.31	36.04
2022	44.92	3.33	3.40	3.32	3.17	41.86

waterbodies, vegetation, or agricultural land, and areas with thermal stress having worst EEI or strongest UHI (UTFVI > 0.020) are either barren lands, industrial areas, or dense urban structures. The analysis of UTFVI revealed an increase in the heat-exposed area from 36.37% (2010) to 40.39% (2019) under the worst category, while the area in the excellent category decreased from 48.43% (2010) to 44.24% (2019). The reason behind this variation in UTFVI could be the increase or decrease in impervious areas and green cover in the study area, which was previously mentioned. During the heatwave, the heat-exposed area (worst category) further increased by 1–2% in 2022 in comparison to 2019 (non-HW condition). Other UTFVI categories have a similar proportion of area, although their order of rank varies. In 2020 and 2021, there was an increase in the excellent category area, which corresponds to the comfort zone, and a decrease in the worst category area, which corresponds to the discomfort zone. The previously discussed COVID-19 lockdown conditions are the cause of this trend.

5 Implications for managing the thermal environment

The interrelationship between LULC, urban thermal environment, and public health highlights the necessity for scientifically informed planning approaches. These approaches directly contribute in achieving Sustainable Development Goals such as Good Health and Well-being (Goal 3), Sustainable Cities and Communities (Goal 11), and Climate Action (Goal 13). The approaches are as follows:

1. The replacement of dark, low-albedo materials with high-reflectance (cool) surfaces has shown significant reduction in surface temperatures by approximately 10–15 °F (Middel et al. 2015).
2. The use of permeable materials enhances evaporative cooling and reduces surface heat accumulation, making this an effective approach for heat mitigation in urban areas.
3. Constructing of energy-efficient and self-cooling buildings with improved natural ventilation as well as use of reflective exteriors and passive cooling technologies may reduce anthropogenic heat emissions.
4. Blue and green infrastructures are essential for thermal regulation through evaporative cooling and shading (Cai et al. 2018). The development of reflecting pools and urban wetlands could improve microclimatic balance. Increasing urban green cover through multi-layered vegetation structures and micro-green solutions such as green roofs and vertical gardens may reduce surface temperature.
5. Promoting public transit networks and reducing private vehicle dependency might indirectly support UHI mitigation.
6. Integrating thermal management strategies such as urban ventilation corridors, green buffers, and open spaces into urban planning may improve air circulation and mitigate localized heating.

6 Conclusion

The study analyses the variations in the land surface temperature, urban heat island effect, and thermal comfort of Delhi, India, and its four major surrounding cities, particularly in relation to land use and land cover changes under different temporal and environmental conditions. The Landsat satellite data is used and processed to derive land surface temperature, from which various thermal comfort indices are calculated, such as the UHI_{index} and UTFVI. The analysis covers multiple time periods, allowing to detect both decadal trends and interannual variations, including conditions influenced by the COVID-19 lockdown and the occurrence of heatwaves.

The positive finding of the study is an increase in vegetation during 2010–2020 due to the creation of biodiversity parks, restriction on mining, and restoration activities. It is found that urban areas have grown significantly over the last decade, which contributes to the increase in the LST. The results indicate a rising trend in LST, and the observed changes in the LST during 2010–2019 were a result of urban expansion, urban geometry, and anthropogenic activities. There is a decrease in LST between 2020 and 2021 due to COVID-19 lockdown conditions, which is caused by a reduction in anthropogenic activities, pollution, and changes in weather conditions. During heatwave conditions, spatial patterns reveal that

there is a wider extent in the area which is affected by thermal discomfort and an intensification of the UHI effect.

This study investigated the presence and extent of urban heat and cool islands and the thermal comfort level in the area using LST. The hotspots are identified over impervious or dry surfaces, whereas the coldspots are found over pervious or moisture-content surfaces. During the HW condition, there is a further increase in the area of higher UHI intensity and a rise in the discomfort zone in comparison to the non-HW condition, which is caused by the synergistic interaction between UHI and heatwave. However, the area under the comfort zone increased during 2020 and 2021 as it shifted to the lower range of the UHI_{index} and UTFVI. There is a need for more green and pervious spaces to improve the thermal environment, enhance urban people's health, and maintain harmony with nature.

The use of comprehensive mitigation measures can limit the increase in UHI and thermal discomfort, such as urban plantscaping, reducing waste heat emissions, use of lighter-colored surfaces, construction of self-cooling buildings and urban wetlands, and raising public awareness about UHI. This study will be helpful for urban planners to maintain the city landscape through better management and planning for the creation of blue-green infrastructure.

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Author contributions **BS** - Conceptualization, Data curation, Methodology, processing and analysis of the data, Simulation, Writing - original draft**VV** - Supervision, Writing - review & editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical responsibilities All authors have read, understood, and complied.

Consent to participate Not applicable.

Consent for publication All of the authors consented to publish this manuscript.

Clinical trial number Not applicable.

Competing interests The authors declare no competing interests.

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